

Control sets that the model has not seen: building held-out domain sets for GoLLeM-v6 and checking them against the training pool

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Abstract

We built five small held-out *control sets* (three English, two Polish) to steer data decisions for GoLLeM-v6 without spending the board test sets, and checked every control document against the 9.39 B-token training pool of our GoLLeM-v5 final runs with a word 13-gram scan and a rule fixed before the scan. The check removed 31 %, 43 % and 49 % of the documents of the three English sets (educational web, Wikipedia, and SQuAD, whose contexts are Wikipedia paragraphs) and under 1 % of the two Polish sets. Without it, 44 % of the English “held-out” documents would have been text the model had trained on. The scan needs only a CPU and about three hours.

1 Why control sets

For the next generation of models we want to measure, during training, how well a model predicts text of several kinds, and to decide about the data from that measurement. The board test sets cannot be used for this: every decision taken on a test set spends part of its credibility. We therefore built five small *control sets*, used only for measurement: never trained on and never published. A control set is useful only if the model has not seen its documents during training; otherwise a low loss measures memory, not generalisation. This note describes how we built the sets and how we checked that they are absent from the training pool.

2 The five sets

Each set is a random sample of whole documents (at least 200 characters, fixed seed 20260926) of about 2.5 M tokens of the GoLLeM-v5 tokenizer, taken from a pinned revision of a public source:

set	kind	source (licence)
en_edu	English, educational web	FineWeb-Edu, one shard of CC-MAIN-2024-10 (ODC-BY)
en_enc	English, encyclopedic	Wikipedia EN, dump 2023-11-01 (CC BY-SA)
en_qa	English, questions and answers	SQuAD v2, train and validation (CC BY-SA 4.0)
pl_enc	Polish, encyclopedic	Wikipedia PL, dump 2023-11-01 (CC BY-SA)
pl_gen	Polish, general web	FineWeb-2 pol_Latn, test split (ODC-BY)

The measure is bits per byte per set (summed negative log-likelihood over UTF-8 bytes, no BOS token, 1024-token windows with reset), which stays comparable when the tokenizer changes. The sets can be rebuilt from the sources above; we do not publish the kept or the removed document lists.

3 Cleaning before the pool check

A document that fails any check is removed whole; nothing is edited inside a document.

*Prepared with the Kolektyw AI agents: Hart (method, scan and draft), Latarnik (editing), Wartownik (review), Monter (pool rebuild).

1. **Benchmarks.** Articles whose titles occur in WikiText-2 (test and validation) are excluded at sampling time. Every document is then scanned with 13-gram and short-question matching against ARC (all splits), BLiMP, WikiText-2 (all splits) and our protected list of evaluation questions, and with a sentence-embedding check (MiniLM, pinned) against the same list.
2. **Personal data.** An automatic flagger for phone numbers, e-mail addresses and Polish identifiers runs on every document; a second, independent reviewer pass removes further documents.

Removed so far (sample $\rightarrow$ kept): **en_edu** 65 (62 flags for personal data and 4 for benchmarks; one document carries two flags), **en_enc** 21 (benchmark 20, personal data 1), **en_qa** 9 (benchmark), **pl_enc** 0, **pl_gen** 97 (personal data). The kept sets have 1 918, 3 484, 6 603, 3 148 and 1 507 documents and 2.22–2.50 M tokens each.

4 The pool check

The training pool of the GoLLeM-v5 final runs has 7 088 917 documents and 9.39 B tokens (sha256 **ecfd0a40**). Looking up every document of the pool in the control sets would be slow; the control sets are small, so we scan in the opposite direction.

1. **Index.** Every control document is normalised (lower case; every character that is not a letter, digit, underscore or whitespace becomes a space) and cut into word 13-grams; each 13-gram is hashed to 8 bytes (BLAKE2b). The index holds the hashes of all control 13-grams.
2. **Scan.** The pool is read once as a token stream, split into segments at the end-of-text token and at the chat end-of-turn token (8 455 478 segments; the pool has 7 088 917 documents, some of which contain chat turns), decoded and normalised the same way. For every 13-gram of a segment that is in the index we count in how many segments it occurs (its document frequency, df). The scan stores only these counts, never text.
3. **Attribution.** Back on the control side, each control document gets the set of its 13-grams found in the pool, with their df, and its coverage: the share of its 13-grams that occur in the pool.

Rule, fixed before the scan (26 September). A control document is removed if it shares at least one 13-gram with the pool whose df is at most 50, *or* if its coverage is at least 0.05. The first condition catches a real copy or a quoted passage; a 13-gram that occurs in more than 50 pool segments is treated as boilerplate (a licence line, a navigation template). The second condition catches text that the pool contains many times (for example Wikipedia mirrors), whose 13-grams all have a high df.

Cost. The scan runs on the CPU only (five worker processes at low priority on a desktop, next to other work) and took 3 h 15 min for the whole pool; no GPU and no cloud machine are needed, and the control sets never leave the machine. A probe on the first 5 M tokens of the pool already found two pool segments sharing 10 distinct 13-grams with the control sets.

5 Results

The scan read all 9 391 706 576 tokens of the pool. 42 557 pool segments share at least one 13-gram with the control sets, and 430 511 distinct control 13-grams occur in the pool. Applying the rule of Section 4:

set	documents before	removed by the pool check	documents kept	tokens kept
en_edu	1 918	589 (30.7 %)	1 329	1 359 813
en_enc	3 484	1 482 (42.5 %)	2 002	1 244 956
en_qa	6 603	3 266 (49.5 %)	3 337	1 256 184
pl_enc	3 148	22 (0.7 %)	3 126	2 433 344
pl_gen	1 507	7 (0.5 %)	1 500	2 210 236

The English sets lose a large share because the pool contains the same kinds of text: Wikipedia, and educational web pages of the FineWeb and DCLM type. `en_qa` loses the most: SQuAD contexts are Wikipedia paragraphs, and the median coverage of its removed documents is 0.34, so these are copies rather than chance overlaps. The pool of the v5 runs is English, so the Polish sets are almost untouched here; they must be checked again against the Polish training pool of v6 before they are used there.

A change made after the scan. Our specification asked for at least 2.0 M tokens per set. After the pool check removed 31–49 % of the English documents, we lowered the English threshold from 2.0 M to 1.2 M tokens (28 September, merge script `a5330407`); the Polish threshold is unchanged. The threshold sets only the size of a measurement set and selects neither a model nor training data. One measurement uses 262 144 tokens per set, so 1.24–1.36 M tokens leave about $4.7\times$ margin.

6 Limitations

A 13-gram match finds copies and long quotations, not paraphrases, translations or the same fact written differently; a control document can be absent from the pool and still cover knowledge the model has learnt. The benchmark check adds a semantic filter, but only against the protected evaluation questions, not against the whole pool. The pool check covers the pool of the final runs; a new pool must be checked again. Normalisation removes punctuation and case, so formatting differences do not hide a copy, but a copy split by inserted text into pieces shorter than 13 words is missed, and so is a 13-gram that crosses a segment boundary of the pool (an end-of-text or end-of-turn token).

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